

Now, we illustrate the overall procedure to find out an aggregated ranking from a list of input rankings.

- **Step 1:** Initially, a similarity matrix between different rankings is computed using the weighted Spearman's footrule similarity measure.
- **Step 2:** From the similarity matrix, the two most similar rankings are chosen and they are merged in order to get more accurate ranking. The merging of two rankings is done by using the corresponding weight of these rankings. This merging technique takes into account the goodness of the parent rankings.

Now if the weight of ranking R_1 is $w(R_1)$ and weight of ranking R_2 is $w(R_2)$, then according to the formula, the merge score $\mathcal{MS}^k$ of the object k is calculated as follows.

$$\mathcal{MS}^k = \frac{w(R_1) * a + w(R_2) * b}{w(R_1) + w(R_2)}, \quad (2)$$

where a, b are the scores of the same object based on the positions in two different rankings. In this way, the merged scores of all the objects are computed. Finally, the ranking can be obtained after sorting all the merged scores. If multiple rankings contain similar values then all the possible merging are computed and suitable ranking with minimum distance to all input ranking is chosen.

- **Step 3:** In this process, the weight of the new ranking is also computed. To compute the weight of new ranking, past experience (goodness of the parent rankings) as well as the present fitness (based on the closeness of new ranking with all the input rankings) are considered.
- **Step 4:** The steps 1-3 are repeated until a single aggregated ranking is reached.

Experimental Results

We have artificially generated several datasets with different dimensions by adding Gaussian noise to each input rank list in a step-wise manner. Inclusion of noise means position of a few bits are altered from the original ranking. When the noise is maximum, most of the bits are altered. For the dataset, the algorithm is applied for 50 iterations and in each iteration the Gaussian noise is incremented slowly by varying the standard deviation. Now, in each iteration (i.e., for any particular step), the rank aggregation algorithm is applied on the input ranking list and the weighted similarity values (Spearman's footrule distance) between the resultant aggregated ranking and the input rankings are computed (see Fig. 1). Finally, to compute the area under the curve (AUC) trapezoidal function is applied. The AUC values for different rank aggregation algorithms and for the proposed approach are given in Table 1.

To investigate the performance of the proposed method in crowdsourcing domain, the method is applied on real-life RTE dataset (?). In this experiment, we have used majority voting as the baseline depending on which the accuracy, specificity, sensitivity and recall of the annotators are computed. Finally, the consensus ranking of them are calculated by aggregating those list of ranks and the final aggregated

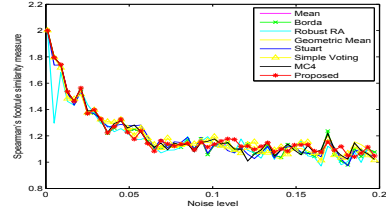


Figure 1: Comparative values of Spearman's footrule similarity measure between the aggregated ranking and all of the input rankings with the increment of noise.

Table 1: The AUC values obtained for the proposed approach and other existing approaches for different instances of a dataset with dimension 20×30 . Here, the standard deviation is varied by 0.004 (Instance 1), 0.01 (Instance 2) and 0.02 (Instance 3) in each step. The best scores over a column (i.e., for a particular instance) are shown in bold.

Algorithm	Instance 1	Instance 2	Instance 3
MC4	0.2360	0.3394	1.0359
Robust RA	0.2301	0.3227	1.0124
Mean	0.2342	0.3378	1.0273
Geometric	0.2356	0.3385	1.0344
Stuart	0.2345	0.3364	1.0314
Simple Voting	0.2363	0.3398	1.0325
Borda	0.2344	0.3379	1.0296
Proposed	0.2367	0.3416	1.0406

Table 2: Accuracy values obtained for different opinion aggregation approaches by applying on the RTE dataset. The best accuracy scores over a column are shown in bold.

Algorithm	RTE Dataset
Majority Voting	89.88%
MACE (?)	93.00%
Raykar (?)	93.00%
GLAD (?)	78.7%
DS (?)	82.2%
Proposed	93.37%

opinion is predicted. Superiority of the scheme in terms of accuracy is reported in Table 2.

Conclusion

In this paper, we show the utility of weighted rank aggregation in the field of crowd based judgment analysis problem. To investigate how good the proposed method is, with respect to other methods, the applications has been done on a real-life AMT dataset. The model can be made more robust if the crowd workers' self-reported confidence scores are taken as input while collecting the annotation scores.

Acknowledgment

The work of Malay Bhattacharyya is supported by the Visvesvaraya Young Faculty Research Fellowship 2015-16 of MeitY, Government of India. All the authors would like to thank the crowd contributors involved in this work.

Dolore modi vel corrupti, consequuntur ipsa eum asper-
natur autem assumenda, autem rem modi cupiditate volup-
tate nulla eveniet dignissimos nesciunt itaque, ea nisi